

Team Listen

Does the team actually listen? A causal audit of language grounding in multi-robot policies

Multi-agent RL (MAPPO)

Isaac Lab

VLM / language conditioning

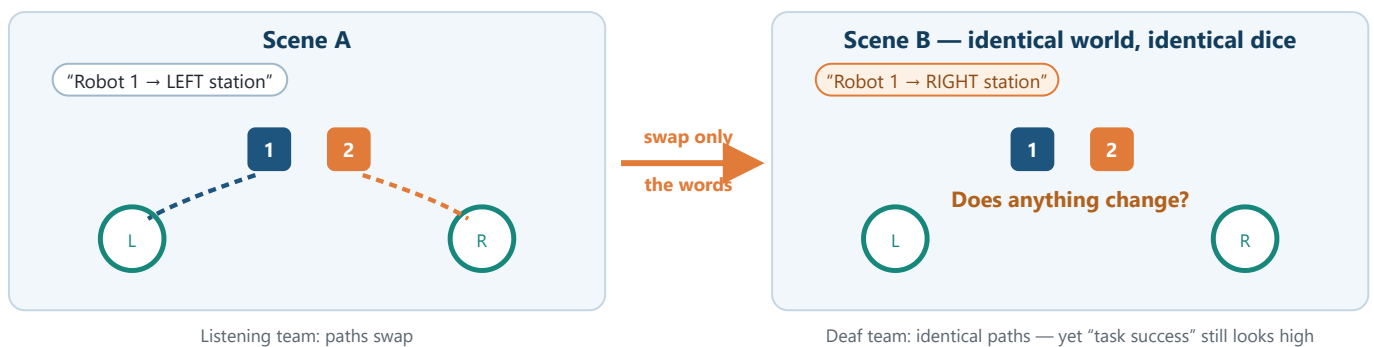
100% simulation

1× RTX 5090

One journal paper, 6–12 months

1 The idea in one picture

Language-commanded robots are the biggest trend in robot learning, and recent audits showed single-arm systems often **ignore the words** and act on what they see. Nobody has checked robot **teams** — where language does work vision cannot: deciding *who* takes *which* identical-looking target, or who goes first. Our audit freezes the world and changes **only the sentence**:



The counterfactual-instruction test. Aggregate success cannot tell these two teams apart; our audit can.

2 What we build — three instruments

Certified tasks

Cooperative Isaac Lab tasks (role binding, precedence, exclusion, reassignment) where geometry alone **cannot** reveal the right assignment — and a machine-checked **certificate** proving it.

The CSI metric

Coordination Sensitivity Index: how much behaviour changes when only the instruction changes, normalised by ordinary run-to-run noise. A deafness meter for robot teams.

Four language interfaces

Broadcast sentence · per-robot prompts · frozen VLM "commander" · sentence embedding — trained identically, audited identically, compared fairly.

Pre-registered headline bet: teams will listen for **role assignment** but be nearly deaf to **precedence** ("robot 1 docks first") — because collision avoidance produces turn-taking for free. A team can obey half its orders, ignore the other half, and still score full marks on every existing benchmark.

3 Why this gap is real — and ours to take

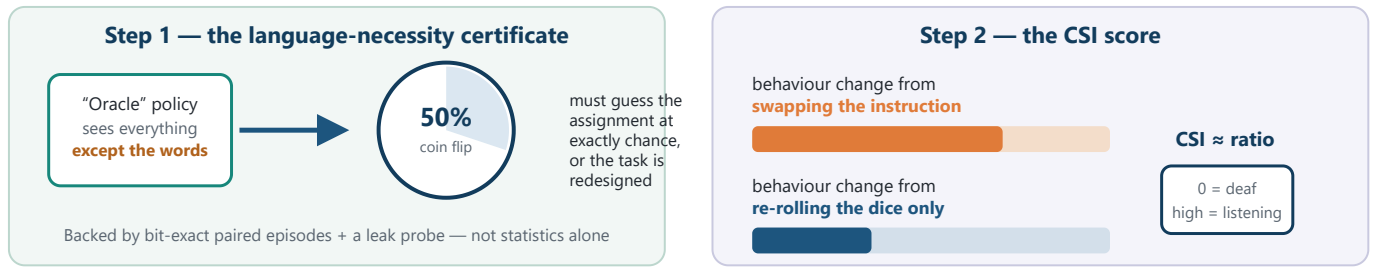
Single-arm audits stop at one robot. LIBERO-Plus / LangGap only perturb object names. Role binding, precedence, exclusion are inexpressible with one arm.

No metric survives team randomness. Raw behaviour deltas are meaningless in MARL; the seed-normalisation in CSI is the missing piece. Verified unoccupied by an adversarial novelty search of 2024–26 work.

Multi-robot language papers never audit. CHORUS, MA-VLA and peers report aggregate success only — the exact setting where a dead language channel hides.

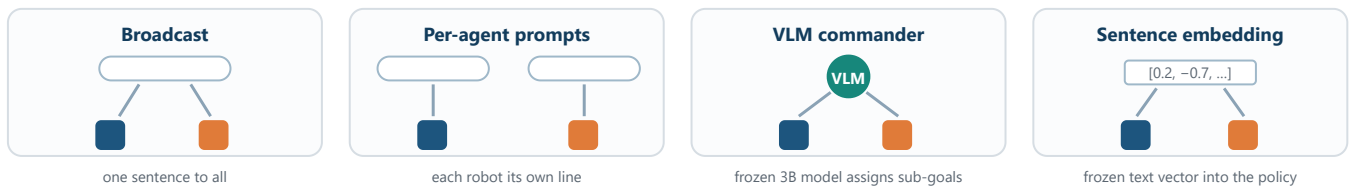
Fits our resources exactly. Everything language-side is frozen (cached embeddings, distilled commander) — only small MARL policies train. One RTX 5090 covers the full program in ≈ 15 GPU-hours.

4 How the audit works



Certificate first (is language even needed?), then CSI (is it actually used?), per semantic category, with clustered bootstrap confidence intervals.

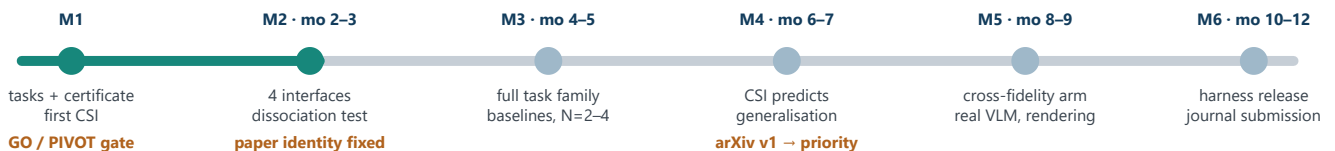
Four ways to give a team its orders — compared head-to-head



Same tasks, same MARL training, same audit — only the language pathway differs. Which interface makes a team actually listen?

5 The paper and the road to it

Venue: IEEE Transactions on Artificial Intelligence (primary; the only IEEE transactions with precedent for evaluation-methodology papers and no hardware expectation, ~12-week first decisions), with the new IEEE T-Robot Learning as a hedged alternative and TCDS → TNNLS → TMLR as the backup chain. An arXiv preprint at month 7 protects priority in a subgenre producing competing preprints monthly. The paper is framed as *measurement methodology*: contributions are the certificate, the CSI metric with formal properties, the dissociation finding, and the open harness.



Twelve-month roadmap. Green = territory already entered. Writing runs alongside: the pre-registration doubles as the first draft of the methods sections.

6 Where we are today

Area	Status
Direction & novelty	✓ Done — survived an adversarial search for prior work; decisions logged in-repo
Design & spec	✓ Done — red-teamed by three independent reviews; 24 critical flaws fixed before any code
Code & tests	✓ Done — environment (bit-parity with our published rendezvous env), rewards, banks, rollout, certificate, CSI — 17 test files, all green
GPU validation	✓ Passed — 390k env-steps/s (3.9× the gate); audit machinery bit-exact on GPU at 4000 parallel envs; full program ≈ 15 GPU-hours
Training	⚙ In progress — a mid-training optimiser instability was root-caused and fixed (LR ceiling); the pre-planned reward-shaping pilot now runs to establish competent learning before the headline campaign
Pre-registration	✓ Frozen — evaluation protocol locked in git before any policy evaluation; OSF registration is the month-1 writing task

7 Risks, honestly

Risk	Answer
"Teams ignore language" sounds pre-announced	The contribution is the instruments + the dissociation, pre-registered; months 2–3 find the differentiated result or pivot early
Policies may fail to learn the tasks	A pre-registered pilot ladder (shaping → simpler grids) runs before anything headline; even a redesign is reportable evidence
Getting scooped	Scope freeze month 6, arXiv month 7; the exact delta was verified unoccupied
Journal rejects simulation-only work	Primary venue chosen for proven tolerance; cross-fidelity arm answers the embodiment question; backup chain pre-agreed